BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Infinite Series

Convergence of Series

NOVEMBER 16, 2025

CONDUCTED BY

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Infinite Series

Infinite Series

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. The infinite series generated by this sequence is the formal expression

$$\sum_{n=1}^{\infty} a_n.$$

Convergence of Infinite Series

Convergence of Infinite Series

We define the corresponding sequence of partial sums (s_m) by

$$s_m = a_1 + a_2 + a_3 + \cdots + a_m,$$

and say that the series $\sum_{n=1}^{\infty} a_n$ converges to S if the sequence (s_m) converges to S. In this case, we write

$$\sum_{n=1}^{\infty} a_n = S.$$

If the sequence $\{S_N\}$ does not converge, the series is called divergent.

Problem

Discuss the convergence or otherwise of the series

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots+\frac{1}{n(n+1)}+\cdots \text{ to } \infty.$$

Problem

Test the convergence or otherwise of the series

$$\sum_{n=0}^{\infty} (-1)^n.$$

Divergence

Problem

Discuss the convergence or divergence of the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

Convergence using MCT

Problem

Test the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Convergence of Geometric Series

Theorem

The geometric series

$$1 + x + x^2 + x^3 + \cdots \text{ to } \infty$$

satisfies the following:

- it converges if $-1 < x < 1$, i.e. if $|x| < 1$;
- it diverges if $x \geq 1$;
- it oscillates finitely if $x = -1$;
- it oscillates infinitely if $x < -1$.

Comparison Test

Comparison Test

Let $\{a_n\}$ and $\{b_n\}$ be sequences of non-negative real numbers such that

$$0 \leq a_n \leq b_n \qquad \text{for all } n.$$

Then:

(i) If $\sum_{n=1}^{\infty} b_n$ converges, then

$$\sum_{n=1}^{\infty} a_n \quad \text{also converges.}$$

(ii) If $\sum_{n=1}^{\infty} a_n$ diverges, then

$$\sum_{n=1}^{\infty} b_n \quad \text{also diverges.}$$

Convergence of p-series

Theorem

The geometric series

$$1 + x + x^2 + x^3 + \cdots \text{ to } \infty$$

satisfies the following:

- it converges if $-1 < x < 1$, i.e. if $|x| < 1$;
- it diverges if $x \geq 1$;
- it oscillates finitely if $x = -1$;
- it oscillates infinitely if $x < -1$.

Thank You!

We'd love your questions and feedback.

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References

[1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.

[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.